

Acoustic Detections from Tidbits Exploratory Analysis

Sander Elliott

```
knitr::opts_chunk$set(warning = T, error = T, message = F, tidy = T)
options(tibble.width = Inf, tibble.print_max = Inf, tibble.print_min = Inf)
```

Load Packages

GLATOS is a great package for acoustic telemetry work. It has to be installed via github. The code for that is commented out. I encourage looking through the glatos package in the help window.

```
#install.packages("remotes")
#remotes::install_github("ocean-tracking-network/glatos")
library(glatos)
?glatos
library(tidyverse)
```

Warning: package 'tidyverse' was built under R version 4.5.2

Warning: package 'ggplot2' was built under R version 4.5.1

Warning: package 'tidyr' was built under R version 4.5.1

Warning: package 'readr' was built under R version 4.5.1

Warning: package 'purrr' was built under R version 4.5.1

Warning: package 'stringr' was built under R version 4.5.1

Warning: package 'forcats' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.1

```
library(sf)
```

Warning: package 'sf' was built under R version 4.5.1

```
library(ggspatial)
```

Warning: package 'ggspatial' was built under R version 4.5.1

```
library(ggmap)
```

Warning: package 'ggmap' was built under R version 4.5.1

```
library(fst)
```

Warning: package 'fst' was built under R version 4.5.2

Load Data

I will load Atlantic sturgeon detections from 2016-2018 as an example. Export criteria are as follows:

Species: AST By Detection Dates Begin Date: 01/01/2016 End Date: 12/31/2018 Tag Type: Acoustic Export: Export Fish Histories Interval: 1 (Raw detections)

Exporting raw detections will take some time. It pulls the full fish history for every fish detected so you may get detections outside of your time frame. If Tidbits won't export try a shorter time frame and combine them later.

Load Raw Detections and write FST (SKIP TO "Load FST")

Load Tidbits download

Raw detections mess up github something fierce, so I will pull directly from my computer

```
df <- read_csv(
  "C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv",
  show_col_types = FALSE)
```

Warning: One or more parsing issues, call `problems()` on your data frame for details,
e.g.:

```
dat <- vroom(...)
problems(dat)
```

```
bad <- problems(df)
bad
```

```
# A tibble: 257 x 5
   row    col expected actual
  <int> <int> <chr>    <chr>
1  877369    18 a double VR2AR-545687
2  877371    18 a double VR2AR-545687
3  877372    18 a double VR2AR-545687
4  877374    18 a double VR2AR-545687
5  877377    18 a double VR2AR-545687
6  877391    18 a double VR2AR-545687
7  879185    18 a double VR2AR-545687
8  879186    18 a double VR2AR-545687
9  879187    18 a double VR2AR-545687
10 879190    18 a double VR2AR-545687
11 879193    18 a double VR2AR-545687
12 879195    18 a double VR2AR-545687
13 879197    18 a double VR2AR-545687
14 1121098    18 a double VR2AR-545681
15 1121105    18 a double VR2AR-545681
16 1121115    18 a double VR2AR-545681
17 1121150    18 a double VR2AR-545681
18 1121151    18 a double VR2AR-545681
19 1121152    18 a double VR2AR-545681
20 1121153    18 a double VR2AR-545681
21 1121155    18 a double VR2AR-545681
22 1121157    18 a double VR2AR-545681
23 1121161    18 a double VR2AR-545681
24 1121162    18 a double VR2AR-545681
25 1121176    18 a double VR2AR-545681
26 1121179    18 a double VR2AR-545681
```

27	1121182	18 a double	VR2AR-545681
28	1121185	18 a double	VR2AR-545681
29	1121194	18 a double	VR2AR-545681
30	1333313	18 a double	VR2AR-545681
31	1333314	18 a double	VR2AR-545681
32	1333316	18 a double	VR2AR-545681
33	1333318	18 a double	VR2AR-545681
34	1333321	18 a double	VR2AR-545681
35	1333322	18 a double	VR2AR-545681
36	1333325	18 a double	VR2AR-545681
37	1333328	18 a double	VR2AR-545681
38	1333329	18 a double	VR2AR-545681
39	1333331	18 a double	VR2AR-545681
40	1333333	18 a double	VR2AR-545681
41	1333335	18 a double	VR2AR-545681
42	1333338	18 a double	VR2AR-545681
43	1333348	18 a double	VR2AR-545681
44	1333349	18 a double	VR2AR-545681
45	1333351	18 a double	VR2AR-545681
46	1333352	18 a double	VR2AR-545681
47	1333353	18 a double	VR2AR-545681
48	1333355	18 a double	VR2AR-545681
49	1333361	18 a double	VR2AR-545681
50	1333365	18 a double	VR2AR-545681
51	1333369	18 a double	VR2AR-545681
52	1333373	18 a double	VR2AR-545681
53	1333375	18 a double	VR2AR-545681
54	1333390	18 a double	VR2AR-545681
55	1333391	18 a double	VR2AR-545681
56	1333392	18 a double	VR2AR-545681
57	1333393	18 a double	VR2AR-545681
58	1333395	18 a double	VR2AR-545681
59	1333396	18 a double	VR2AR-545681
60	1333398	18 a double	VR2AR-545681
61	1333400	18 a double	VR2AR-545681
62	1333401	18 a double	VR2AR-545681
63	1333402	18 a double	VR2AR-545681
64	1333403	18 a double	VR2AR-545681
65	1333405	18 a double	VR2AR-545681
66	1333406	18 a double	VR2AR-545681
67	1333408	18 a double	VR2AR-545681
68	1333410	18 a double	VR2AR-545681
69	1333412	18 a double	VR2AR-545681

70	1333414	18 a double	VR2AR-545681
71	1333416	18 a double	VR2AR-545681
72	1333418	18 a double	VR2AR-545681
73	1333420	18 a double	VR2AR-545681
74	1333422	18 a double	VR2AR-545681
75	1333423	18 a double	VR2AR-545681
76	1333425	18 a double	VR2AR-545681
77	1333429	18 a double	VR2AR-545681
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81	1333469	18 a double	VR2AR-545681
82	1333470	18 a double	VR2AR-545681
83	1333471	18 a double	VR2AR-545681
84	1333472	18 a double	VR2AR-545681
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93	1333485	18 a double	VR2AR-545681
94	1333495	18 a double	VR2AR-545681
95	1333497	18 a double	VR2AR-545681
96	1333498	18 a double	VR2AR-545681
97	1333499	18 a double	VR2AR-545681
98	1333500	18 a double	VR2AR-545681
99	1333501	18 a double	VR2AR-545681
100	1333502	18 a double	VR2AR-545681
101	1333503	18 a double	VR2AR-545681
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109	1333511	18 a double	VR2AR-545681
110	1333512	18 a double	VR2AR-545681
111	1333514	18 a double	VR2AR-545681
112	1333517	18 a double	VR2AR-545681

113	1333519	18	a double	VR2AR-545681
114	1333520	18	a double	VR2AR-545681
115	1333521	18	a double	VR2AR-545681
116	1333522	18	a double	VR2AR-545681
117	1333523	18	a double	VR2AR-545681
118	1333524	18	a double	VR2AR-545681
119	1333532	18	a double	VR2AR-545681
120	1333540	18	a double	VR2AR-545681
121	1333558	18	a double	VR2AR-545681
122	1333561	18	a double	VR2AR-545681
123	1333562	18	a double	VR2AR-545681
124	1333565	18	a double	VR2AR-545681
125	1333566	18	a double	VR2AR-545681
126	1333567	18	a double	VR2AR-545681
127	1333569	18	a double	VR2AR-545681
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135	1333594	18	a double	VR2AR-545681
136	1333595	18	a double	VR2AR-545681
137	1333596	18	a double	VR2AR-545681
138	1333599	18	a double	VR2AR-545681
139	1333601	18	a double	VR2AR-545681
140	1333603	18	a double	VR2AR-545681
141	1333605	18	a double	VR2AR-545681
142	1333630	18	a double	VR2AR-545681
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144	1333633	18	a double	VR2AR-545681
145	1333635	18	a double	VR2AR-545681
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147	1333641	18	a double	VR2AR-545681
148	1333643	18	a double	VR2AR-545681
149	1333653	18	a double	VR2AR-545681
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151	1333661	18	a double	VR2AR-545681
152	1333662	18	a double	VR2AR-545681
153	1333663	18	a double	VR2AR-545681
154	1333664	18	a double	VR2AR-545681
155	1333666	18	a double	VR2AR-545681

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157	1333669	18 a double	VR2AR-545681
158	1333671	18 a double	VR2AR-545681
159	1333673	18 a double	VR2AR-545681
160	1333674	18 a double	VR2AR-545681
161	1333676	18 a double	VR2AR-545681
162	1333677	18 a double	VR2AR-545681
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177	1333706	18 a double	VR2AR-545681
178	1333708	18 a double	VR2AR-545681
179	1333709	18 a double	VR2AR-545681
180	1333711	18 a double	VR2AR-545681
181	1333712	18 a double	VR2AR-545681
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183	1333714	18 a double	VR2AR-545681
184	1333715	18 a double	VR2AR-545681
185	1333716	18 a double	VR2AR-545681
186	1333717	18 a double	VR2AR-545681
187	1333718	18 a double	VR2AR-545681
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189	1333721	18 a double	VR2AR-545681
190	1333724	18 a double	VR2AR-545681
191	1333727	18 a double	VR2AR-545681
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193	1333731	18 a double	VR2AR-545681
194	1333732	18 a double	VR2AR-545681
195	1333733	18 a double	VR2AR-545681
196	1333736	18 a double	VR2AR-545681
197	1333737	18 a double	VR2AR-545681
198	1333739	18 a double	VR2AR-545681

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200	1434086	18 a double	VR2AR-545687
201	1434093	18 a double	VR2AR-545687
202	1434099	18 a double	VR2AR-545687
203	1434101	18 a double	VR2AR-545687
204	1434104	18 a double	VR2AR-545687
205	1434105	18 a double	VR2AR-545687
206	1434106	18 a double	VR2AR-545687
207	1434107	18 a double	VR2AR-545687
208	1434108	18 a double	VR2AR-545687
209	1434109	18 a double	VR2AR-545687
210	1434115	18 a double	VR2AR-545687
211	1434117	18 a double	VR2AR-545687
212	1434120	18 a double	VR2AR-545687
213	1434124	18 a double	VR2AR-545687
214	1434125	18 a double	VR2AR-545687
215	1682043	18 a double	VR2AR-545687
216	1682045	18 a double	VR2AR-545687
217	1682046	18 a double	VR2AR-545687
218	1682048	18 a double	VR2AR-545687
219	1682050	18 a double	VR2AR-545687
220	1682052	18 a double	VR2AR-545687
221	1682053	18 a double	VR2AR-545687
222	1682056	18 a double	VR2AR-545687
223	1682057	18 a double	VR2AR-545687
224	1682059	18 a double	VR2AR-545687
225	1682060	18 a double	VR2AR-545687
226	1682061	18 a double	VR2AR-545687
227	1682062	18 a double	VR2AR-545687
228	1682063	18 a double	VR2AR-545687
229	1682065	18 a double	VR2AR-545687
230	1690184	18 a double	VR2AR-545687
231	1690187	18 a double	VR2AR-545687
232	1690189	18 a double	VR2AR-545687
233	1690190	18 a double	VR2AR-545687
234	1690202	18 a double	VR2AR-545687
235	1690207	18 a double	VR2AR-545687
236	1690211	18 a double	VR2AR-545687
237	1765604	18 a double	VR2AR-545687
238	1765610	18 a double	VR2AR-545687
239	1765613	18 a double	VR2AR-545687
240	1765614	18 a double	VR2AR-545687
241	1908844	18 a double	VR2AR-545687

242 1908846 18 a double VR2AR-545687
243 1908850 18 a double VR2AR-545687
244 1908852 18 a double VR2AR-545687
245 2064695 18 a double VR2AR-545687
246 2064697 18 a double VR2AR-545687
247 2064698 18 a double VR2AR-545687
248 2177108 18 a double VR2AR-545687
249 2177109 18 a double VR2AR-545687
250 2177111 18 a double VR2AR-545687
251 2177112 18 a double VR2AR-545687
252 2177138 18 a double VR2AR-545687
253 2177149 18 a double VR2AR-545687
254 2177177 18 a double VR2AR-545687
255 2177180 18 a double VR2AR-545687
256 2177182 18 a double VR2AR-545687
257 2177183 18 a double VR2AR-545687

file

<chr>

1 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
2 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
3 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
4 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
5 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
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18 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
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20 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
21 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
22 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
23 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
24 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
25 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```

241 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
242 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
243 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
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250 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
251 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
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253 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
254 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
255 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
256 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv
257 C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv

```

```

bad_rows <- df[bad$row, ]
unique(bad$col)

```

```
[1] 18
```

```
unique(bad$actual)
```

```
[1] "VR2AR-545687" "VR2AR-545681"
```

```
unique(bad_rows[,18])
```

```

# A tibble: 7 x 1
  RxID
  <dbl>
1 547818
2      NA
3 545680
4 545683
5 545685
6 547822
7 547816

```

```
rm(df); gc()
```

	used (Mb)	gc trigger (Mb)	max used (Mb)
Ncells	1460964 78.1	2386848 127.5	2386848 127.5
Vcells	2526566 19.3	47925689 365.7	57066338 435.4

Re-try df with rxid as character

```
df2 <- read_csv(
  "C:/Users/sander.elliott/Downloads/tidbits-export-1770136798454.csv",
  col_types = cols(
    RxID = col_character()
  )
)

bad2 <- problems(df2)
bad2
```

```
# A tibble: 0 x 5
# i 5 variables: row <int>, col <int>, expected <chr>, actual <chr>, file <chr>
```

```
unique(df2[,18])
```

```
# A tibble: 220 x 1
  RxID
  <chr>
1 <NA>
2 100306
3 100304
4 100307
5 100308
6 100309
7 100310
8 100311
9 100312
10 100316
11 100319
```


12 100322
13 101373
14 100323
15 100879
16 103145
17 103153
18 104104
19 115088
20 106545
21 106531
22 125892
23 101377
24 125889
25 109568
26 125899
27 115091
28 110837
29 480044
30 480040
31 115060
32 115064
33 101366
34 115070
35 125888
36 115069
37 105585
38 104679
39 100313
40 103142
41 103141
42 103143
43 103146
44 103148
45 103150
46 103152
47 104107
48 115089
49 123258
50 101362
51 101372
52 101365
53 101385
54 104678

55 101364
56 106543
57 106528
58 106540
59 104684
60 1938
61 2628
62 1947
63 2672
64 1949
65 2666
66 5070
67 5076
68 1957
69 2627
70 5071
71 5075
72 125896
73 106542
74 125891
75 115081
76 106538
77 101376
78 101371
79 105586
80 106550
81 104686
82 106541
83 106534
84 106530
85 104685
86 101368
87 5077
88 1632
89 5081
90 5074
91 106535
92 109251
93 110836
94 109246
95 110834
96 110828
97 110428

98 109243
99 105583
100 107726
101 104106
102 106539
103 101386
104 104681
105 101381
106 105584
107 101357
108 104677
109 110833
110 119982
111 125895
112 125894
113 115087
114 100303
115 119642
116 115092
117 106533
118 115059
119 101383
120 115086
121 547817
122 547819
123 547821
124 547818
125 545686
126 547822
127 545685
128 104105
129 115079
130 115072
131 111480
132 115078
133 106547
134 121010
135 101379
136 108902
137 120980
138 108900
139 109248
140 108924

141 110831
142 108796
143 125500
144 108913
145 109254
146 108903
147 104011
148 VR2AR-545687
149 545680
150 547816
151 547820
152 480047
153 480043
154 108920
155 135056
156 108904
157 108798
158 108807
159 108791
160 545687
161 121006
162 109244
163 135053
164 119981
165 110474
166 545682
167 552666
168 552665
169 102400
170 108794
171 108916
172 135055
173 119984
174 135054
175 108806
176 119980
177 108808
178 553493
179 553492
180 552668
181 552667
182 108802
183 109249

```
184 108809
185 554109
186 554107
187 554106
188 554101
189 135057
190 115090
191 121017
192 115062
193 110830
194 109250
195 120978
196 107724
197 121009
198 138582
199 553494
200 554103
201 108919
202 VR2AR-545681
203 135058
204 115084
205 480045
206 550003
207 121016
208 121012
209 108799
210 545683
211 104111
212 107381
213 121018
214 108921
215 120984
216 554108
217 108793
218 108792
219 138586
220 138583
```

```
glimpse(df2)
```

```
Rows: 2,287,429
Columns: 26
```

```

$ Period          <dtm> 2009-06-16 00:00:00, 2010-04-02 06:53:19, 2010-04-
02~
$ tagtype         <chr> "Acoustic", "Acoustic", "Acoustic", "Acoustic", "Acou~
$ Species        <chr> "AST", "AST", "AST", "AST", "AST", "AST", "AST", "AST~
$ FishID         <chr> "AST-2009-001", "AST-2009-001", "AST-2009-001", "AST-
~
$ TagId          <dbl> 1074907, 1074907, 1074907, 1074907, 1074907, 1074907,~
$ Event          <chr> "Initial Capture/Release (Acoustic tag)", "Detection"~
$ pings          <dbl> NA, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ FirstTS        <dtm> 2009-06-16 00:00:00, 2010-04-02 06:53:19, 2010-04-
02~
$ LastTS         <dtm> 2009-06-16 00:00:00, 2010-04-02 06:53:19, 2010-04-
02~
$ SensorType     <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ avgSensorValue <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ avgComputedValue <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ Codespace      <chr> "A69-1303", "A69-1303", "A69-1303", "A69-1303", "A69-
~
$ IDCode         <dbl> 58597, 58597, 58597, 58597, 58597, 58597, 58597, 5859~
$ Frequency      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ avgPower       <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ AntennaID      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ RxID           <chr> NA, "100306", "100306", "100306", "100306", "100306",~
$ DeploymentUID  <dbl> NA, 47, 47, 47, 47, 47, 47, 47, 47, 47, 47, 47, 47, 4~
$ Easting        <dbl> NA, 513573, 513573, 513573, 513573, 513573, 513573, 5~
$ Northing       <dbl> NA, 4939351, 4939351, 4939351, 4939351, 4939351, 4939~
$ LocationCode   <chr> "Penobscot River", "CHPMN", "CHPMN", "CHPMN", "CHPMN"~
$ RKM            <dbl> NA, 12.72, 12.72, 12.72, 12.72, 12.72, 12.72, 12.72, ~
$ UTMZone        <dbl> NA, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 1~
$ SiteCode       <chr> NA, "CHPMN", "CHPMN", "CHPMN", "CHPMN", "CHPMN", "CHP~
$ AltFishID      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~

```

Write FST to save space and remove huge df

```
dim(df2)
```

```
[1] 2287429      26
```

```
names(df2)
```

[1]	"Period"	"tagtype"	"Species"	"FishID"
[5]	"TagId"	"Event"	"pings"	"FirstTS"
[9]	"LastTS"	"SensorType"	"avgSensorValue"	"avgComputedValue"
[13]	"Codespace"	"IDCode"	"Frequency"	"avgPower"
[17]	"AntennaID"	"RxID"	"DeploymentUID"	"Easting"
[21]	"Northing"	"LocationCode"	"RKM"	"UTMZone"
[25]	"SiteCode"	"AltFishID"		

```
names(df2)[1:10]
```

[1]	"Period"	"tagtype"	"Species"	"FishID"	"TagId"
[6]	"Event"	"pings"	"FirstTS"	"LastTS"	"SensorType"

```
str(df2$Period) # or whatever your time column is
```

```
POSIXct[1:2287429], format: "2009-06-16 00:00:00" "2010-04-02 06:53:19" "2010-04-02 06:55:14" ...
```

```
str(df2$Period)
```

```
POSIXct[1:2287429], format: "2009-06-16 00:00:00" "2010-04-02 06:53:19" "2010-04-02 06:55:14" ...
```

```
head(df2$Period, 10)
```

[1]	"2009-06-16 00:00:00 UTC"	"2010-04-02 06:53:19 UTC"
[3]	"2010-04-02 06:55:14 UTC"	"2010-04-02 06:58:27 UTC"
[5]	"2010-04-02 06:59:19 UTC"	"2010-04-02 07:03:20 UTC"
[7]	"2010-04-02 07:04:49 UTC"	"2010-04-02 07:05:45 UTC"
[9]	"2010-04-02 07:06:52 UTC"	"2010-04-02 07:08:22 UTC"

```
# optional drop
df2 <- df2 %>% select(-any_of(c("SensorType", "avgSensorValue", "avgCompute",
                                "Codespace", "AltFishID", "avgComputedValue",
                                "Frequency", "avgPower", "AntennaID"))))

write_fst(df2, "data/ASTdet16to18.fst")
```

Warning: package 'fstcore' was built under R version 4.5.2

```
rm(df2); gc()
```

```
      used (Mb) gc trigger      (Mb) max used   (Mb)
Ncells 1483944 79.3   2386848 127.5   2386848 127.5
Vcells 2579361 19.7   59735730 455.8  63450427 484.1
```

```
rm(list = ls())
rm(list = ls(all.names = TRUE))
```

Load FST

```
det <- read.fst("data/ASTdet16to18.fst")
glimpse(det)
```

Rows: 2,287,429

Columns: 18

```
$ Period      <dtm> 2009-06-16 00:00:00, 2010-04-02 06:53:19, 2010-04-
02 06~
$ tagtype     <chr> "Acoustic", "Acoustic", "Acoustic", "Acoustic", "Acousti~
$ Species     <chr> "AST", "AST", "AST", "AST", "AST", "AST", "AST", "AST", ~
$ FishID      <chr> "AST-2009-001", "AST-2009-001", "AST-2009-001", "AST-
200~
$ TagId       <dbl> 1074907, 1074907, 1074907, 1074907, 1074907, 1074907, 10~
$ Event       <chr> "Initial Capture/Release (Acoustic tag)", "Detection", "~
$ pings       <dbl> NA, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ FirstTS     <dtm> 2009-06-16 00:00:00, 2010-04-02 06:53:19, 2010-04-
02 06~
$ LastTS      <dtm> 2009-06-16 00:00:00, 2010-04-02 06:53:19, 2010-04-
02 06~
$ IDCode      <dbl> 58597, 58597, 58597, 58597, 58597, 58597, 58597, 58597, ~
$ RxID        <chr> NA, "100306", "100306", "100306", "100306", "100306", "1~
$ DeploymentUID <dbl> NA, 47, 47, 47, 47, 47, 47, 47, 47, 47, 47, 47, 47, 47, ~
$ Easting     <dbl> NA, 513573, 513573, 513573, 513573, 513573, 513573, 5135~
$ Northing    <dbl> NA, 4939351, 4939351, 4939351, 4939351, 4939351, 4939351~
$ LocationCode <chr> "Penobscot River", "CHPMN", "CHPMN", "CHPMN", "CHPMN", "~
$ RKM         <dbl> NA, 12.72, 12.72, 12.72, 12.72, 12.72, 12.72, 12.72, 12.~
$ UTMZone     <dbl> NA, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, ~
$ SiteCode    <chr> NA, "CHPMN", "CHPMN", "CHPMN", "CHPMN", "CHPMN", "CHPMN"~
```


Get to Work Fixing up Data

```
detglat <- det %>%
  mutate(YR = year(Period)) %>%
  filter(Event == "Detection",
         YR == "2016" |
         YR == "2017" |
         YR == "2018") %>% ### the years I want to look at RN
  mutate(detection_timestamp_utc = as.POSIXct(Period),
         transmitter_id = as.character(IDCode),
         transmitter_codespace = as.character(TagId),
         receiver_sn = RxID) %>%
  false_detections(tf=3600) %>% ## Remove false detentions based on 1 hr
  filter(., passed_filter == 1) %>%
  st_as_sf(coords = c("Easting", "Northing"), crs = 26919) %>% # NAD83 / UTM zone 19N
  st_transform(4269) %>%
  mutate(
    deploy_long = st_coordinates(.)[, 1],
    deploy_lat = st_coordinates(.)[, 2],
    animal_id = FishID) %>%
  st_drop_geometry()

glimpse(detglat)
```

Rows: 477,868

Columns: 26

\$ Period	<dtm> 2016-04-10 15:49:02, 2016-04-10 15:50:44, 201~
\$ tagtype	<chr> "Acoustic", "Acoustic", "Acoustic", "Acoustic"~
\$ Species	<chr> "AST", "AST", "AST", "AST", "AST", "AST", "AST"~
\$ FishID	<chr> "AST-2009-001", "AST-2009-001", "AST-2009-001"~
\$ TagId	<dbl> 1074907, 1074907, 1074907, 1074907, 1074907, 1~
\$ Event	<chr> "Detection", "Detection", "Detection", "Detect~
\$ pings	<dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
\$ FirstTS	<dtm> 2016-04-10 15:49:02, 2016-04-10 15:50:44, 201~
\$ LastTS	<dtm> 2016-04-10 15:49:02, 2016-04-10 15:50:44, 201~
\$ IDCode	<dbl> 58597, 58597, 58597, 58597, 58597, 58597, 5859~
\$ RxID	<chr> "125889", "125889", "125889", "125889", "12588~
\$ DeploymentUID	<dbl> 3877, 3877, 3877, 3877, 3877, 3877, 3877, 3877~

```

$ LocationCode      <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
$ RKM               <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA~
$ UTMZone           <dbl> 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19, 19~
$ SiteCode          <chr> "WP01", "WP01", "WP01", "WP01", "WP01", "WP01", "WP01"~
$ YR                <dbl> 2016, 2016, 2016, 2016, 2016, 2016, 2016, 2016, 2016~
$ detection_timestamp_utc <dtm> 2016-04-10 15:49:02, 2016-04-10 15:50:44, 201~
$ transmitter_id    <chr> "58597", "58597", "58597", "58597", "58597", "~
$ transmitter_codespace <chr> "1074907", "1074907", "1074907", "1074907", "1~
$ receiver_sn       <chr> "125889", "125889", "125889", "125889", "12588~
$ min_lag           <dbl> 102, 79, 79, 107, 90, 57, 57, 71, 69, 69, 95, ~
$ passed_filter     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ deploy_long       <dbl> -68.83152, -68.83152, -68.83152, -68.83152, -
6~
$ deploy_lat        <dbl> 44.45549, 44.45549, 44.45549, 44.45549, 44.455~
$ animal_id         <chr> "AST-2009-001", "AST-2009-001", "AST-2009-
001"~

```

get detection events & summarize

```

detev <- detection_events(detglat, location_col = "SiteCode")
glimpse(detev)

```

```

Rows: 63,559
Columns: 9
$ event      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16,~
$ animal_id  <chr> "AST-2009-001", "AST-2009-001", "AST-2009-001", "AST-
2~
$ location   <chr> "WP01", "FP01", "OR01", "FK01", "HARR03", "FK01", "FK0~
$ mean_latitude <dbl> 44.45549, 44.46911, 44.54256, 44.55729, 44.58609, 44.5~
$ mean_longitude <dbl> -68.83152, -68.80665, -68.74894, -68.80599, -68.81270,~
$ first_detection <dtm> 2016-04-10 15:49:02, 2016-04-10 16:52:03, 2016-04-
10 ~
$ last_detection <dtm> 2016-04-10 16:09:00, 2016-04-10 17:51:55, 2016-04-
10 ~
$ num_detections <int> 14, 22, 9, 2, 8, 2, 1, 1, 1, 13, 17, 1, 1, 1, 1, 1,~
$ res_time_sec  <dbl> 1198, 3592, 1022, 52, 2634, 44, 0, 0, 0, 13376, 8037, ~

```

```

detsum <- summarize_detections(detglat, location_col = "SiteCode")
detsum

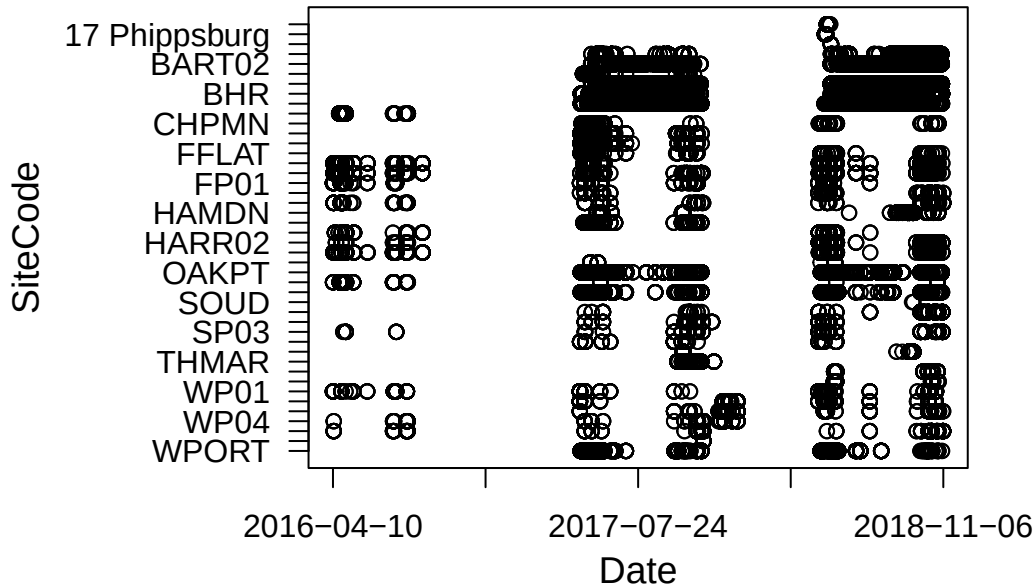
```

	animal_id	num_locs	num_dets		first_det		last_det
1	AST-2009-001	31	50717	2016-04-10	15:49:02	2017-10-22	02:50:24
2	AST-2009-004	31	7243	2016-04-11	05:37:15	2017-05-31	09:56:11
3	AST-2009-007	32	21956	2016-06-02	09:34:24	2017-10-09	07:29:52
4	AST-2009-016	32	34840	2016-05-11	14:03:04	2018-09-11	07:41:01
5	AST-2009-023	24	30885	2017-05-03	05:42:04	2017-07-23	00:25:04
6	AST-2010-001	24	46274	2017-05-29	02:40:40	2017-10-24	04:58:36
7	AST-2010-003	34	61980	2016-07-16	22:56:35	2018-10-19	05:05:12
8	AST-2010-011	13	4833	2016-05-07	03:17:44	2017-10-19	16:23:17
9	AST-2014-003	31	15271	2016-04-23	12:41:11	2017-07-05	15:44:28
10	AST-2017-001	29	26600	2017-10-03	05:54:48	2018-10-30	12:23:42
11	AST-2017-002	28	37400	2017-10-23	09:57:12	2018-10-02	23:38:06
12	AST-2017-004	33	39383	2017-10-29	03:38:10	2018-11-06	04:59:54
13	AST-2017-006	30	34797	2017-11-15	04:15:04	2018-10-26	06:57:10
14	AST-2017-008	31	64704	2017-09-17	04:54:37	2018-10-07	05:10:46
15	AST-2018-005	7	66	2018-10-26	20:54:54	2018-10-27	04:45:42
16	AST-2018-006	8	58	2018-10-25	05:34:05	2018-10-25	14:55:48
17	AST-2018-007	11	179	2018-10-08	13:52:59	2018-10-21	16:20:13
18	AST-2018-008	6	80	2018-10-23	08:25:10	2018-10-23	21:35:57
19	AST-2018-010	12	602	2018-09-25	09:20:27	2018-10-26	07:02:26

1	BART01 BART02 BDNPT BHCov BHR BLEdG CHPMN DPEAST DPWEST FFLAT FK01 FK02 FP01
2	BART01 BART02 BDNPT BHCov BHR BLEdG Buoy F CHPMN DPEAST DPWEST FFLAT FK01 FK02
3	BART01 BART02 BDNPT BHCov BHR BLEdG CHPMN DPEAST DPWEST FFLAT FK01 FK02 FP01
4	BART01 BART02 BDNPT BHCov BHR BLEdG CHPMN DPEAST DPWEST FFLAT FK01 FK02 FP01 FP02
5	BART01 BART02 BDNPT BHCov BHR BLEdG CHPMN DPEAST DPWEST FFLAT FK01 FK02 FP01
6	BART01 BART02 BDNPT BHCov BHR BLEdG CHPMN DPEAST DPWEST FFLAT FK01 FK02 FP01
7	BART01 BART02 BDNPT BHCov BHR BLEdG CHPMN DPEAST DPWEST FFLAT FK01 FK02 FP01 FP02 FP03
8	
9	BART01 BART02 BDNPT BHCov BHR BLEdG Buoy F CHPMN DPEAST DPWEST FFLAT FK01 FK02
10	BART02 BHCov BHR BLEdG CHPMN FFLAT FK01 FK02 FP02 FP03 HARRIS
11	16 Sasanoa 17 Phippsburg 18 Mill Cove BHR BLEdG CHPMN FFLAT FK01 FK02 FP01
12	BART01 BART02 BHCov BHR BLEdG CHPMN FFLAT FK01 FK02 FP01 FP02 FP03 HAMDEN HARRIS
13	BART01 BART02 BHCov BHR BLEdG CHPMN FFLAT FK01 FK02 FP01 FP02 FP03
14	BART01 BART02 BHCov BHR BLEdG CHPMN FFLAT FK01 FK02 FP01 FP02 FP03 HARRIS
15	
16	
17	
18	
19	

GLATOS Plots

```
abacus_plot(detglat, location_col = "SiteCode")
```



```
detection_bubble_plot(detglat, location_col = "SiteCode",
                      background_ylim = c(min(detglat$deploy_lat) - 0.1,
                                          max(detglat$deploy_lat + 0.1)),
                      background_xlim = c(min(detglat$deploy_long) - 0.1)
```

```
Error in parse(text = input): <text>:5:0: unexpected end of input
3:                                     max(detglat$deploy_lat + 0.1)),
4:                                     background_xlim = c(min(detglat$deploy_long) - 0.1)
~
```